

5.3 d -Diagrams

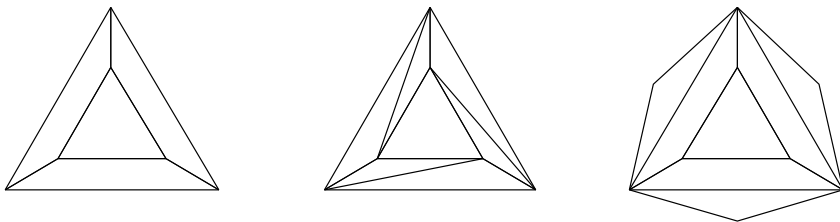
Schlegel diagrams were studied extensively around the turn of this century; however, no one realized that there is a problem: not everything that “looks like” a Schlegel diagram actually is one. (See the notes below.) Now we define *looks like*.

Definition 5.7. A d -*diagram* is a polytopal subdivision $\mathcal{D}$ of a d -polytope, $P = |\mathcal{D}| \subseteq \mathbb{R}^d$, such that $G \cap \partial P$ is a face of P for each $G \in \mathcal{D}$.

A d -diagram $\mathcal{D}$ is *simplicial* if $|\mathcal{D}|$ and all the polytopes in $\mathcal{D}$ are simplices. The diagram is *simple* if every vertex of $|\mathcal{D}|$ is contained in exactly d different d -polytopes of $\mathcal{D}$, and every other vertex of $\mathcal{D}$ is contained in exactly $d + 1$ different d -polytopes in $\mathcal{D}$.

Check that every Schlegel diagram of a d -polytope is a $(d - 1)$ -diagram! We define the face poset $L(\mathcal{D})$, and combinatorial equivalence, of d -diagrams as a special case of the definitions for polyhedral complexes. With this, a Schlegel $(d - 1)$ -diagram $\mathcal{D}(P, F)$ is simplicial, respectively simple (according to Definition 5.7), if and only if the polytope P is simplicial, respectively simple. In fact, a diagram $\mathcal{D}$ is simplicial if and only if its face lattice $\hat{L}(\mathcal{D})$ “looks like” the face lattice of a simplicial polytope, in that all the lower intervals $[\emptyset, G]$ are boolean for $G \neq \hat{1}$; similarly, $\mathcal{D}$ is simple if and only if all upper intervals $[G, \hat{1}]$ are boolean for $G \neq \emptyset$.

Here are a few trivial examples: a Schlegel diagram (of a triangular prism), a 2-diagram that is not a Schlegel diagram, and a polytopal subdivision that is not a 2-diagram.



We note that the question of whether a given d -diagram is a Schlegel diagram can be reduced to a linear programming problem (Exercise 5.2) — but the question of whether it is *combinatorially equivalent to a Schlegel diagram* is much harder; the task can be split into the enumeration problem for oriented matroid spheres, and the oriented matroid realizability problem (see the notes at the end of this lecture). So, the following is a decidedly nontrivial theorem, accompanied by a trivial proof.

Theorem 5.8.

Every 2-diagram is combinatorially equivalent to a Schlegel diagram.

Proof. A simple combinatorial argument implies that the graph of every 2-diagram is 3-connected (it is simple and planar by construction). Thus the theorem follows from Steinitz' Theorem 4.1, together with the construction of Schlegel diagrams of 3-polytopes. $\square$

Proposition 5.9. *If a diagram $\mathcal{D}$ is Schlegel, then $\mathcal{D}$ is a regular subdivision of $F = |\mathcal{D}|$. The converse is true in the case where F is a simplex, but not in general.*

We omit the proof — see Exercises 5.2(ii) and 5.7.

There are other properties that separate Schlegel diagrams from some non-Schlegel diagrams. For example, let $\mathcal{D}$ be a d -diagram, and let L be the “reconstructed” face lattice of the $(d+1)$ -polytope P , if it exists. The combinatorial information about $\mathcal{D}$ is then contained in the pair (L, F) , where F is a distinguished coatom of L . Now we call a diagram *invertible* if for every coatom F' of L , there is a d -diagram corresponding to the pair (L, F') . We know that every Schlegel diagram is invertible. It turns out that a part of the non-Schlegel diagrams is not invertible.

Similarly, we say that $\mathcal{D}$ has a *combinatorial polar* if there is a diagram whose combinatorial data are given by the pair (L^{op}, A) , where A is a vertex of $\mathcal{D}$ (corresponding to an atom of L , and thus to a coatom of L^{op}). Again, every Schlegel diagram has a polar diagram (which is Schlegel), and some non-Schlegel diagrams have polars, some do not.

5.4 Three Examples

Examples of interesting Schlegel diagrams, and of 3-diagrams that are not Schlegel diagrams, are not too hard to come by. In the following we will describe three strikingly simple examples, where two are due to Barnette [47], and one is from Schulz [487]. See also Ewald [201, Sect. IV.4] for the “classical” examples of Brückner and Barnette (see the notes), and Schulz [488] for some other interesting constructions.

From general theory (namely the technique of “Gale diagrams,” which we will soon get to) one can see that every d -diagram with at most $d+4$ vertices is always Schlegel. Therefore, the minimal counterexamples that we can hope for are 3-diagrams with 8 vertices.

Here we go: the following construction produces a 3-diagram that is not combinatorially equivalent to the Schlegel diagram of any 4-polytope.

Example 5.10 (Schulz' 3-diagram). [487]

We start with a 3-polytope Q with 6 vertices, labeled $1, 2, \dots, 6$ in our drawing. The 3-polytope can be realized by starting with a tetrahedron, cutting off the top to get a triangular prism, and then cutting off an extra triangle $[2, 3, 5]$ as in the figure, where $3, 5$ were vertices of the truncated prism, and 2 was on an edge of it.